

Memory Matching Flutter Game

README Documentation

Student Information

- **Student Name & Id:** Ahmed Omae Ali - 1338099
- **Course:** CS5450 Mobile Programming
- **Project:** Exercise 2 – Memory Matching Flutter Game
- **Technology:** Flutter / Dart
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1. Project Overview

This project is a Memory Matching Game developed using Flutter and Dart. The application allows users to flip cards, match image pairs, track scores, and restart the game. The application was successfully tested on a Samsung Android device, Android Emulator, and Chrome browser. The application starts with a welcome screen, then allows users to play a memory matching game by flipping and matching image cards.

2. Features Implemented

- 4 × 5 memory matching board
- 20 animated cards
- Score tracking system
- Move counter
- Restart game button
- Winning dialog popup
- Responsive user interface
- Random colorful card design
- Smooth card animations
- Android device support
- Flutter/Dart implementation

3. Project Structure

```
memory_match_game/
├── assets/
│   └── images/
│       ├── cat.png
│       ├── dog.png
│       ├── lion.png
│       ├── fox.png
│       ├── rabbit.png
│       ├── owl.png
│       ├── tiger.png
│       ├── panda.png
│       ├── monkey.png
│       └── horse.png
├── lib/
│   ├── data/
│   │   └── card_data.dart
│   ├── models/
│   │   └── card_model.dart
│   ├── screens/
│   │   ├── welcome_screen.dart
│   │   └── home_screen.dart
│   ├── widgets/
│   │   └── game_card.dart
│   └── main.dart
└── pubspec.yaml
```

4. Technologies Used

Technology	Purpose
Flutter	Cross-platform UI framework
Dart	Programming language
Android Emulator	Application testing
Samsung Android Device	Real device testing
IntelliJ IDEA	Development IDE

5. How to Configure and Run the Project

Step 1: Install Flutter SDK

Install Flutter SDK from the official Flutter website.

Step 2: Open the Project

Open the project folder using IntelliJ IDEA or Android Studio.

Step 3: Install Dependencies

Run the following command inside the project folder:

```
flutter pub get
```

Step 4: Connect Android Device or Emulator

Connect a physical Android phone or launch an Android emulator.

Step 5: Run the Application

Run the following command:

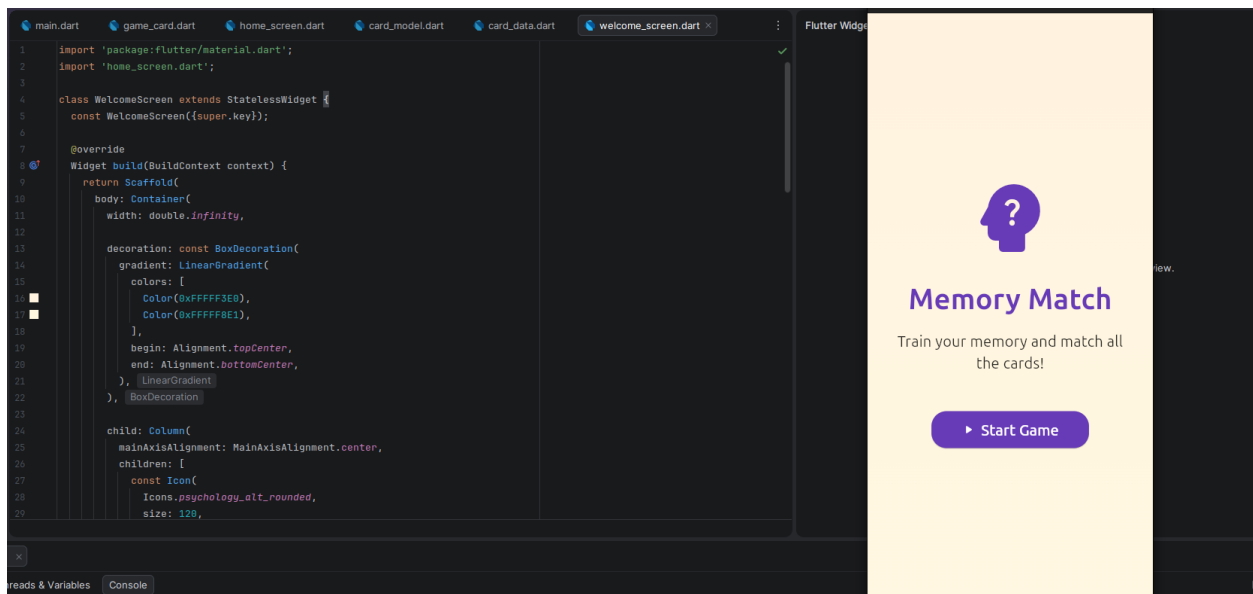
flutter run

Then select the Android device when prompted.

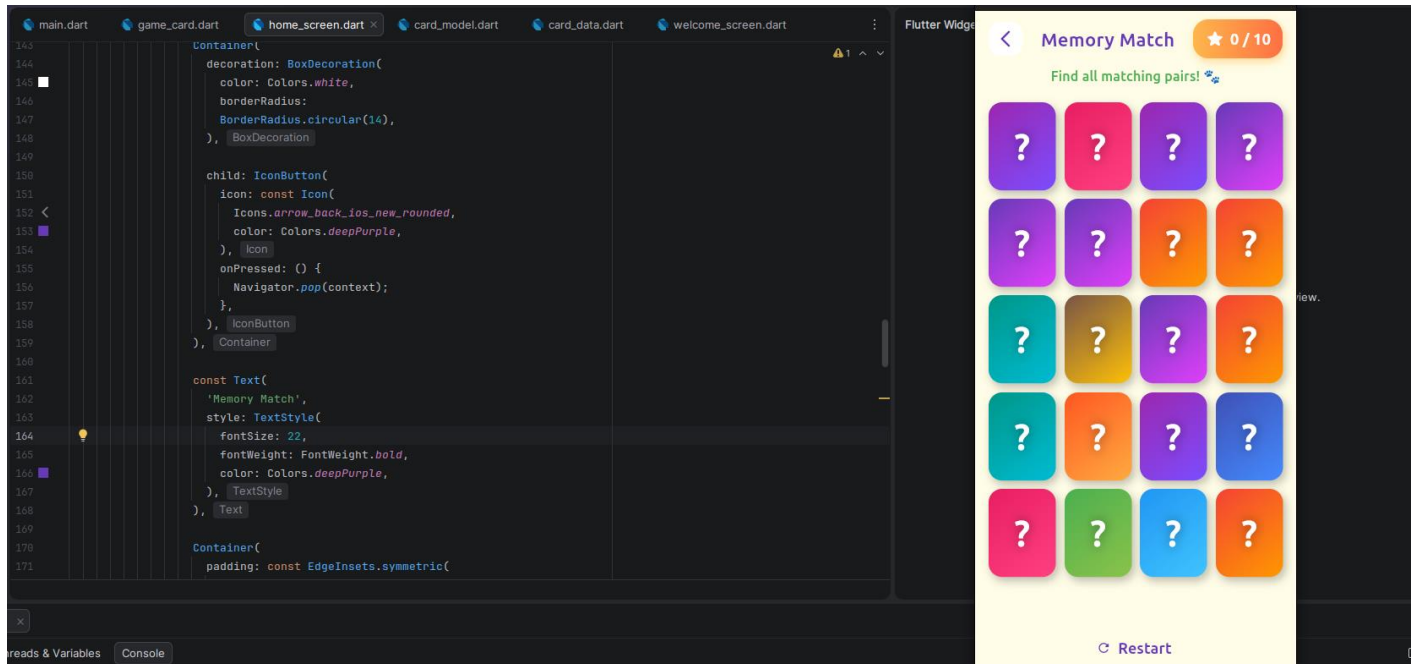
6. Screenshots

Add screenshots here before final submission:

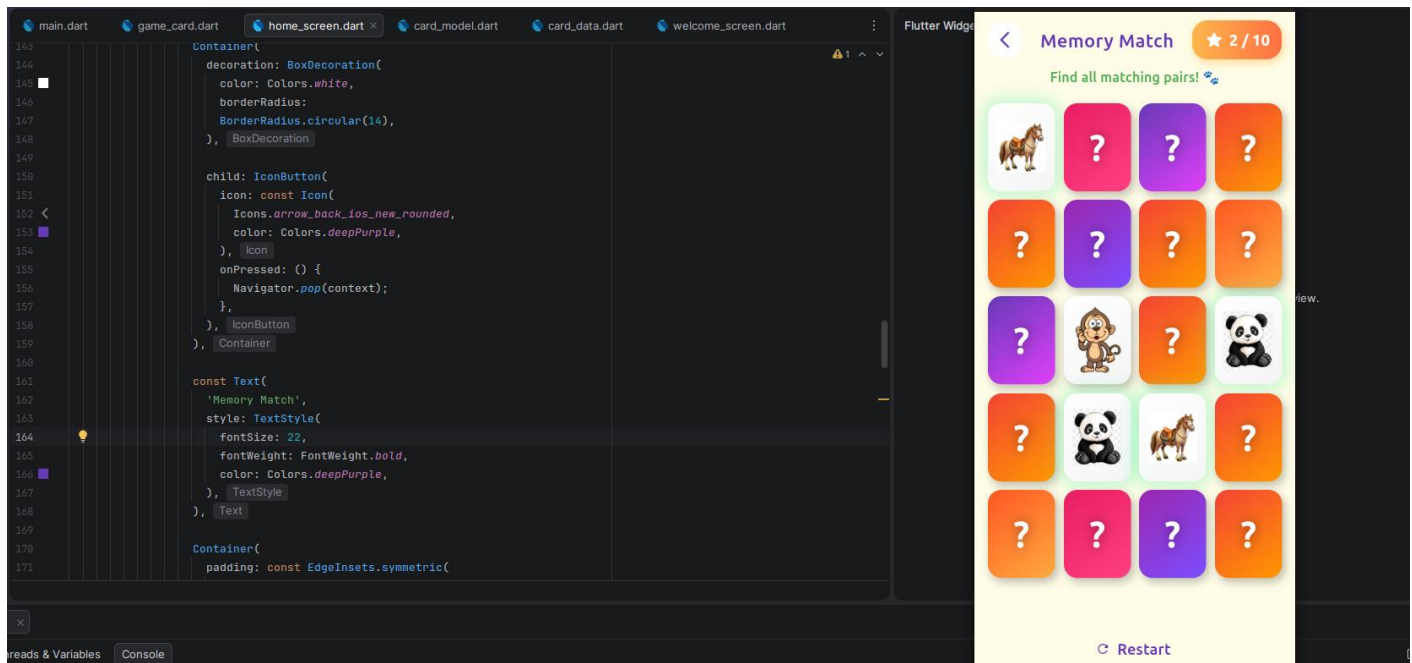
Welcome Screen

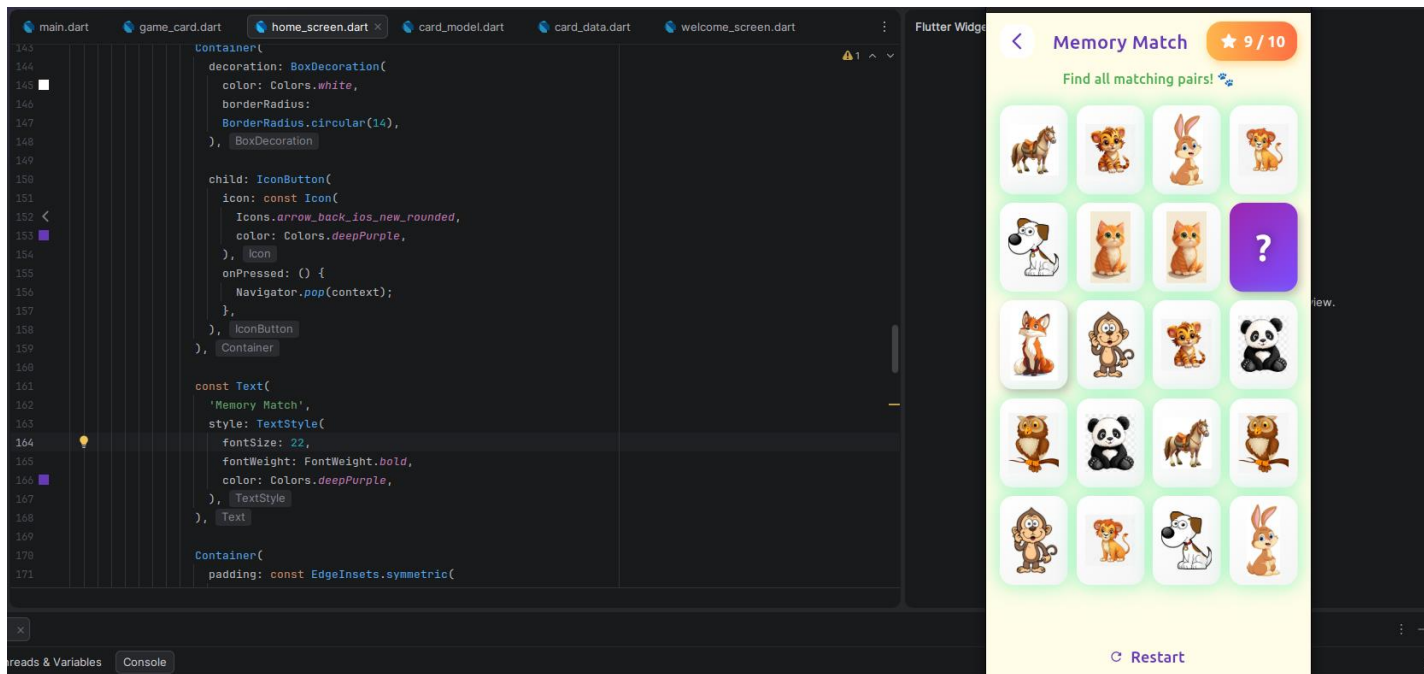


Main Game Screen

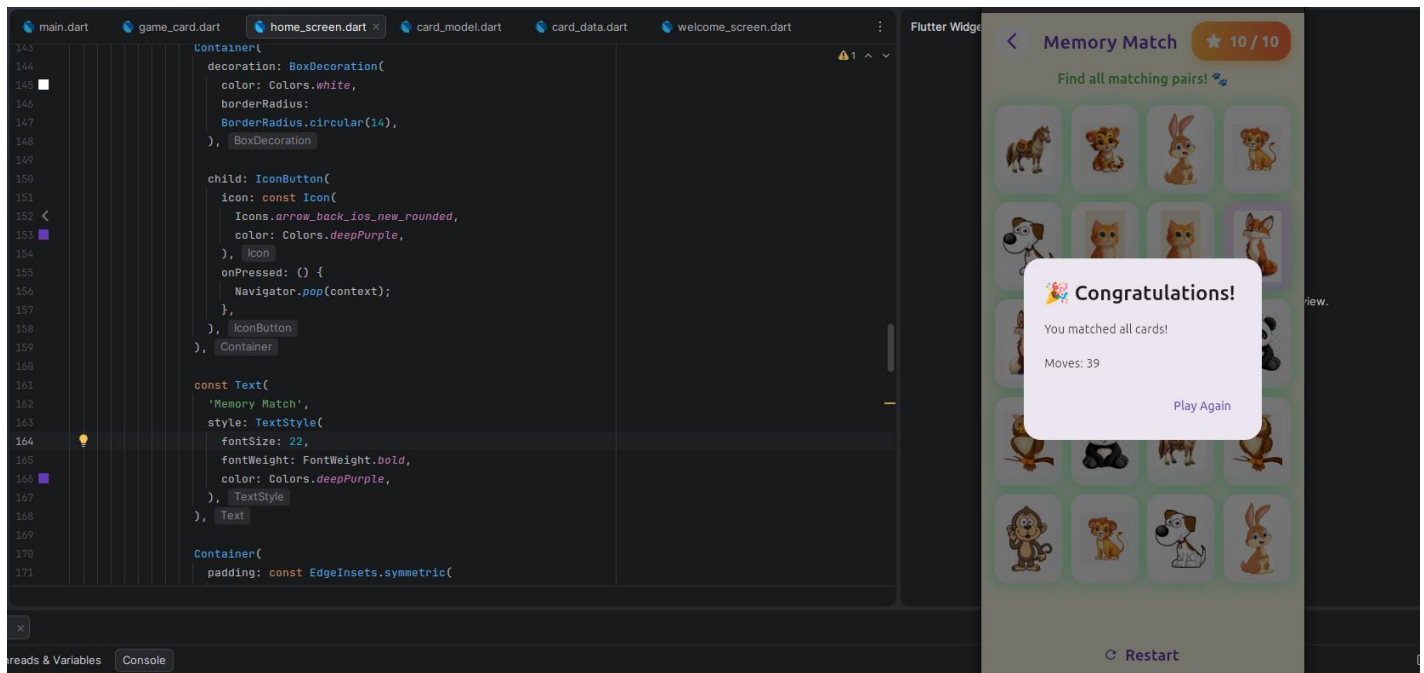


Matching Cards

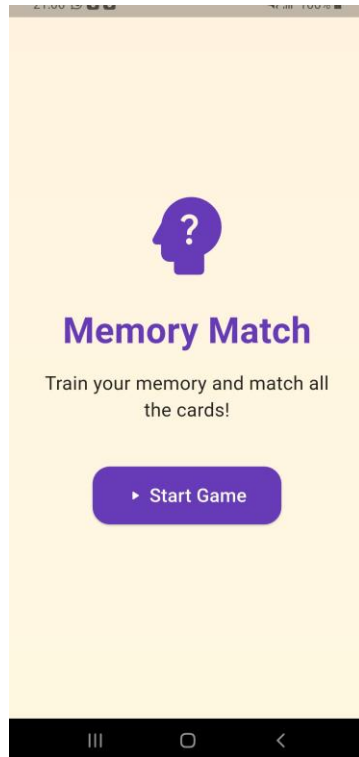


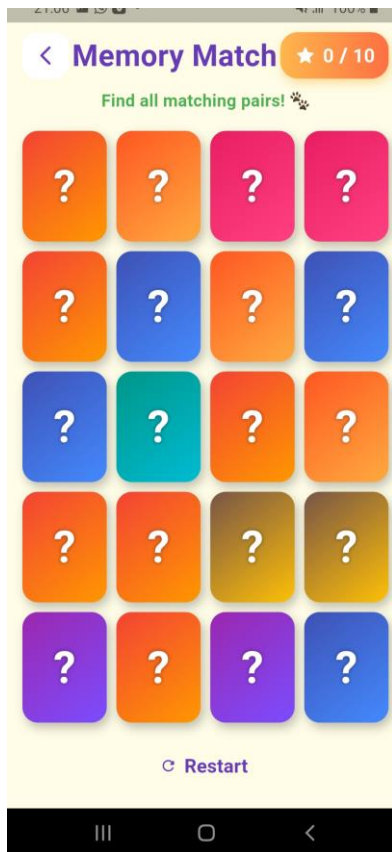


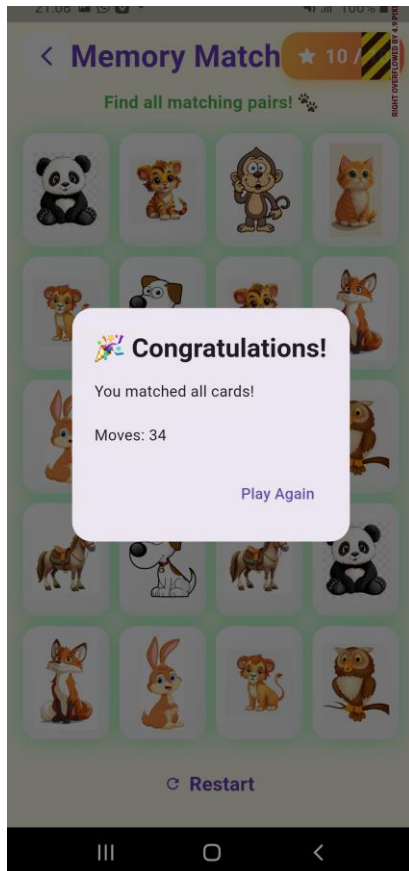
Winning Dialog



Running on Samsung Android Device







7.Challenges Faced

During development, some emulator and Flutter SDK configuration issues were resolved. The final application was optimized for responsive Android device support.

8. Conclusion

This project demonstrates Flutter mobile application development using Dart. The game includes responsive UI design, animations, score tracking, and Android deployment. The application was tested successfully on a real Samsung Android device.